Credit Card Default Dataset - Variable Guide

AAI-500 Project Reference Guide

This guide explains the variables in the Default of Credit Card Clients dataset. The dataset contains customer characteristics, credit limits, six months of repayment status, six months of bill statement amounts, six months of payment amounts, and a binary default outcome.

1. Customer Characteristics

Variable	Meaning
LIMIT_BAL	Amount of the customer's granted credit limit, in New Taiwan dollars (NT$).
SEX	Customer sex code: 1 = male; 2 = female.
EDUCATION	Education code. Documented categories include 1 = graduate school, 2 = university, 3 = high school
MARRIAGE	Marital-status code. Documented categories include 1 = married, 2 = single, and 3 = other.
AGE	Customer age in years.

2. Repayment Status: PAY_0 through PAY_6

PAY variables are repayment-status codes, not dollar amounts. They describe whether the customer's payment was on time or delayed during each month.

Variable	Month represented	Meaning
PAY_0	September 2005	Repayment status
PAY_2	August 2005	Repayment status
PAY_3	July 2005	Repayment status
PAY_4	June 2005	Repayment status
PAY_5	May 2005	Repayment status
PAY_6	April 2005	Repayment status

The UCI documentation explicitly describes -1 as paid duly and positive values 1 through 9 as increasing numbers of months of payment delay. The supplied data also contain values such as -2 and 0 that are not fully defined in the UCI variable description, so those values should not be assigned an invented meaning in the project report. Also note that the original dataset uses PAY_0 and then PAY_2; there is no PAY_1 column.

3. Bill Statement Amounts: BILL_AMT1 through BILL_AMT6

These variables contain the customer's bill statement amounts, measured in NT$. They describe how much was billed, not how much the customer paid.

Variable	Month represented
BILL_AMT1	September 2005
BILL_AMT2	August 2005
BILL_AMT3	July 2005
BILL_AMT4	June 2005
BILL_AMT5	May 2005
BILL_AMT6	April 2005

Example: BILL_AMT1 = 50,000 means the customer's September 2005 bill statement amount was NT$50,000. There is no BILL_0 variable in this dataset.

4. Previous Payment Amounts: PAY_AMT1 through PAY_AMT6

These variables contain the amount of the customer's previous payments, measured in NT$. This is different from the PAY_* variables, which describe repayment status.

Variable	Month represented
PAY_AMT1	September 2005
PAY_AMT2	August 2005
PAY_AMT3	July 2005
PAY_AMT4	June 2005
PAY_AMT5	May 2005
PAY_AMT6	April 2005

5. Target Variable: DEFAULT

Value	Meaning
0	Customer did not default.
1	Customer defaulted.

6. Quick Way to Remember the Similar Names

PAY_0 ... PAY_6 = repayment status (on-time/delayed behavior)
BILL_AMT1 ... BILL_AMT6 = amount on the bill statement
PAY_AMT1 ... PAY_AMT6 = amount actually paid
DEFAULT = whether the customer defaulted

Example Customer

Suppose a customer has LIMIT_BAL = 100,000, PAY_0 = 2, BILL_AMT1 = 50,000, and PAY_AMT1 = 5,000. This means the customer has a NT$100,000 credit limit, a September bill statement of NT$50,000, a previous payment amount of NT$5,000, and a repayment-status code of 2, which the documentation describes as a two-month payment delay.

Project Interpretation

Conceptually, the dataset uses customer characteristics, credit limits, and six months of payment behavior to study credit-card default. For modeling, these variables can be treated as predictors while DEFAULT is the outcome variable. Relationships found by statistical or machine-learning models should be described as associations or predictive relationships, not as proof that a variable causes default.

Source

UCI Machine Learning Repository, Default of Credit Card Clients dataset. Dataset documentation: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients